Dilute Fermi gases: Upper bounds via cluster expansion*

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Many-body quantum system with N particles in large box $\Lambda = [0, L]^3$ and Hamiltonian

$$H_N = \sum_{j=1}^{N} -\Delta_{x_j} + \sum_{1 \le i < j \le N} v(x_i - x_j).$$

- Kinetic energy: Laplacian $-\Delta_{x_j}$
- Interaction energy: Pairwise interaction $v \ge 0$, radial and of compact support.

We consider

- Spin-polarized Fermi gas: Realize H_N on $L^2_a(\Lambda^N) = \bigwedge^N L^2(\Lambda)$
- Spin-$\frac{1}{2}$ Fermi gas: Realize H_N on $L^2_a(\Lambda^{N_\uparrow}) \otimes L^2_a(\Lambda^{N_\downarrow})$

Interested in ground state energy in the thermodynamic limit

$$e(\rho) = \lim_{\substack{L \to \infty \\ N/L^3 \to \rho}} \inf_{\psi \in L^2_a(\Lambda^N)} \frac{\langle \psi | H_N | \psi \rangle \,/\, \langle \psi | \psi \rangle}{L^3},$$

$$e(\rho_\uparrow, \rho_\downarrow) = \lim_{\substack{L \to \infty \\ N_\sigma / L^3 \to \rho_\sigma}} \inf_{\psi \in L^2_a(\Lambda^{N_\uparrow}) \otimes L^2_a(\Lambda^{N_\downarrow})} \frac{\langle \psi | H_N | \psi \rangle \,/\, \langle \psi | \psi \rangle}{L^3}.$$

Dilute limit $\rightsquigarrow$ ρ small compared to scattering length of v (i.e. $a^3\rho \ll 1$)

Definition

The s- and p-wave scattering lengths a_s and a_p are defined by

$$4\pi a_s = \inf\left\{\int \left(|\nabla f|^2 + \frac{1}{2}vf^2\right) \mathrm{d}x : f(x) \to 1 \text{ for } |x| \to \infty\right\},$$

$$12\pi a_p^3 = \inf\left\{\int \left(|\nabla f|^2 + \frac{1}{2}vf^2\right) |x|^2 \,\mathrm{d}x : f(x) \to 1 \text{ for } |x| \to \infty\right\}.$$

The minimizing f's are the s- and p-wave scattering functions. They are denoted f_{s0} and f_{p0} respectively.

Study minimal energy of two particles in a box with

- no symmetry (bosons/fermions of different spin) $\rightsquigarrow$ s-wave,
- fermionic symmetry (fermions of same spin) $\rightsquigarrow$ p-wave.

p-wave in 3 dimensions $\simeq$ s-wave in 5 dimensions. For $|x| >$ range of v

$$f_{s0}(x) = 1 - \frac{a_s}{|x|}, \qquad f_{p0}(x) = 1 - \frac{a_p^3}{|x|^3}.$$

Bosons

Realize $H_N = \sum_{j=1}^N -\Delta_{x_j} + \sum_{1\le i<j\le N} v(x_i - x_j)$ on $L^2_s(\Lambda^N) = L^2(\Lambda)^{\otimes_{\mathrm{sym}} N}$.

Theorem (Dyson 1957; Lieb–Yngvason 1998; Yau–Yin 2009; Fournais–Solovej 2020 (+ many others))

For appropriate v we have

$$e_{\mathrm{Bose}}(\rho) = 4\pi a_s \rho^2 \left[1 + \frac{128}{15\sqrt{\pi}}(a_s^3\rho)^{1/2} + o((a_s^3\rho)^{1/2})\right], \qquad a_s^3\rho \ll 1.$$

- Energy of one pair of particles is $8\pi a_s/L^3$. (Short calculation)
- There are $N(N-1)/2 \simeq N^2/2$ many pairs.

$\rightsquigarrow$ Energy density $4\pi a_s \rho^2$.

Goal for fermions: Find "analogue of $4\pi a_s \rho^2$"

Spin-1/2 Fermi gas

Theorem (Lieb–Seiringer–Solovej 2005)

Let $v \geq 0$ be radial and of compact support. Then for $a_s^3\rho \ll 1$

$$e(\rho_\uparrow, \rho_\downarrow) = \frac{3}{5}(6\pi^2)^{2/3}\left(\rho_\uparrow^{5/3} + \rho_\downarrow^{5/3}\right) + 8\pi a_s \rho_\uparrow \rho_\downarrow + a_s \rho^2 \delta(a_s^3\rho),$$

with $\rho = \rho_\uparrow + \rho_\downarrow$ and $-C(a_s^3\rho)^{1/39} \leq \delta(a_s^3\rho) \leq C(a_s^3\rho)^{2/27}$.

Kinetic energy $\frac{3}{5}(6\pi^2)^{2/3}\left(\rho_\uparrow^{5/3} + \rho_\downarrow^{5/3}\right)$

Interaction energy $8\pi a_s \rho_\uparrow \rho_\downarrow$. Similar to $4\pi a_s \rho^2$ for the Bose gas:

- Energy of one pair of spin-$\uparrow$ and spin-$\downarrow$ is $8\pi a_s / L^3$. (Short calculation)
- There are $N_\uparrow N_\downarrow$ many such pairs.

$\rightsquigarrow$ Energy density $8\pi a_s \rho_\uparrow \rho_\downarrow$.

Smallest energy (for fixed ρ) for $\rho_\uparrow = \rho_\downarrow = \rho/2$.

$$e(\rho_\uparrow, \rho_\downarrow) = \frac{3}{5}(6\pi^2)^{2/3}\left(\rho_\uparrow^{5/3} + \rho_\downarrow^{5/3}\right) + 8\pi a_s \rho_\uparrow \rho_\downarrow + a_s \rho^2 \delta(a_s^3 \rho).$$

Expect the **Huang-Yang term** for $\rho_\uparrow = \rho_\downarrow = \rho/2$

$$\delta(a_s^3\rho) = c_{\mathrm{HY}} a_s \rho^{1/3} + o(a_s \rho^{1/3}), \qquad c_{\mathrm{HY}} = \frac{4(11 - 2\log 2)}{35\pi^2}\left(\frac{3}{4\pi}\right)^{4/3}$$

$\rightsquigarrow$ "Optimal" error is $\delta(a_s^3\rho) = O((a_s^3\rho)^{1/3})$.

$$e(\rho_\uparrow, \rho_\downarrow) = \frac{3}{5}(6\pi^2)^{2/3}\left(\rho_\uparrow^{5/3} + \rho_\downarrow^{5/3}\right) + 8\pi a_s \rho_\uparrow \rho_\downarrow + a_s \rho^2 \delta(a_s^3 \rho).$$

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$\rightsquigarrow$ "Optimal" error is $\delta(a_s^3\rho) = O((a_s^3\rho)^{1/3})$.

Bounds on the error: (Assume always $v \geq 0$ radial and compactly supported.)

$-(a_s^3\rho)^{1/39} \lesssim \delta(a_s^3\rho) \lesssim (a_s^3\rho)^{2/27}$	(Lieb–Seiringer–Solovej 2005)
$-(a_s^3\rho)^{1/9} \lesssim \delta(a_s^3\rho) \lesssim (a_s^3\rho)^{2/9}$	v also smooth (Falconi–Giacomelli–Hainzl–Porta 2021)
$-(a_s^3\rho)^{1/5} \lesssim \delta(a_s^3\rho) \lesssim (a_s^3\rho)^{1/3}$	v also smooth (Giacomelli 2022)
$\delta(a_s^3\rho) \lesssim_\varepsilon (a_s^3\rho)^{1/3-\varepsilon}$ for any $\varepsilon > 0$	(L. 2023)

Spin-polarized Fermi gas

Theorem (L.–Seiringer 2023)

Let $v \geq 0$ be radial and of compact support. Then for $a_p^3 \rho \ll 1$

$$e(\rho) \leq \frac{3}{5}(6\pi^2)^{2/3}\rho^{5/3} + \frac{12\pi}{5}(6\pi)^{2/3} a_p^3 \rho^{8/3}\left[1 + 0 a_p \rho^{1/3} - \frac{9}{35}(6\pi^2)^{2/3} a_0^2 \rho^{2/3}(1+o(1))\right].$$

(Analogous results in dimensions $d = 1, 2$.)

$$3a_0^2 := \frac{1}{12\pi a_p^3}\int_{\mathbb{R}^3} |x|^4 \left(|\nabla f_{p0}(x)|^2 + \frac{1}{2}v(x)|f_{p0}(x)|^2\right) \mathrm{d}x,$$

v allowed to have a hard core $v(x) = +\infty$ for $|x| \leq r_0$.

Expected from physics literature with $=$ (Efimov–Amus'ya 1965; Efimov 1966; Wellenhofer–Drischler–Schwenk 2020; ...)

$$e(\rho) = \frac{3}{5}(6\pi^2)^{2/3}\rho^{5/3} + \frac{12\pi}{5}(6\pi)^{2/3} a_p^3 \rho^{8/3} + 0 a_p^4 \rho^3 + C_{10/3}\rho^{10/3} + \dots$$

1st order perturbation theory – Naive $a^3\rho^{8/3}$

Consider trial state ψ_{FFG} of Slater determinant with momenta in $B_F = \{k \in \frac{2\pi}{L}\mathbb{Z}^3 : |k| \leq k_F\}$ the Fermi ball, $k_F \simeq (6\pi^2)^{1/3}\rho^{1/3}$. Then

$$\langle \psi_{\mathrm{FFG}} | H_N | \psi_{\mathrm{FFG}} \rangle = \sum_{k \in B_F} |k|^2 + \frac{1}{2} \iint \rho^{(2)}(x_1, x_2) v(x_1 - x_2) \, \mathrm{d}x_1 \, \mathrm{d}x_2$$

Taylor $\rho^{(2)} \simeq \frac{(6\pi^2)^{2/3}}{5} \rho^{8/3} |x_1 - x_2|^2$

$$\simeq \frac{3}{5}(6\pi^2)^{2/3} L^3 \rho^{5/3} + \frac{(6\pi^2)^{2/3}}{10} L^3 \rho^{8/3} \int |x|^2 v(x) \, \mathrm{d}x$$

Pretend $24\pi a_p^3 \simeq \int |x|^2 v(x) \, \mathrm{d}x$ (valid for smooth and small v.)

$$\simeq \frac{3}{5}(6\pi^2)^{2/3} L^3 \rho^{5/3} + \frac{12\pi}{5}(6\pi^2)^{2/3} L^3 a_p^3 \rho^{8/3}$$

Need to include interactions in trial state $\rightsquigarrow$ include the scattering functions f_{p0} in trial state.

We consider Jastrow-type trial state

$$\psi_N(x_1,\dots,x_N) = \frac{1}{\sqrt{C_N}} \prod_{1\le i<j\le N} f_p(x_i - x_j) D_N(x_1,\dots,x_N)$$

f_p scaled and cut-off version of f_{p0}, D_N Slater determinant, C_N normalization constant.

Compute energy

$$\begin{aligned}&\langle \psi_N | H_N | \psi_N \rangle \\ &= E_0 + \iint \rho^{(2)}_{\mathrm{Jas}}(x_1,x_2) \left(\left| \frac{\nabla f(x_1-x_2)}{f(x_1-x_2)} \right|^2 + \frac{1}{2} v(x_1-x_2) \right) \mathrm{d}x_1\, \mathrm{d}x_2 \\ &\qquad + \text{3-body term}\end{aligned}$$

E_0 kinetic energy of Slater determinant, $\rho^{(2)}_{\mathrm{Jas}}$ the 2-particle density. Think $\rho^{(2)}_{\mathrm{Jas}}(x_1,x_2) \simeq f(x_1-x_2)^2 \rho^{(2)}(x_1,x_2)$.

Compute $\rho^{(2)}_{\mathrm{Jas}} \rightsquigarrow$ **Cluster expansion**.

Positive temperature

At positive temperature, consider pressure

$$\psi(\beta,\mu) = \lim_{L\to\infty} \sup_{\Gamma} \frac{\frac{1}{\beta}S(\Gamma) - \mathrm{Tr}_{\mathcal{F}}[(\mathcal{H}-\mu\mathcal{N})\Gamma]}{L^3}.$$

Here

$$\mathcal{H} = (H_n)_{n\in\mathbb{N}}, \qquad \mathcal{N} = (n)_{n\in\mathbb{N}}$$

on Fock spaces

$$\mathcal{F} = \bigoplus_{n=0}^{\infty} \bigwedge^{n} L^2(\Lambda;\mathbb{C}^2) \quad \text{spin-1/2}$$

$$\mathcal{F} = \bigoplus_{n=0}^{\infty} \bigwedge^{n} L^2(\Lambda) \qquad \text{spin-polarized}$$

Pressure should be compared to pressure of free gas

$$\psi_0(\beta,\mu) = \frac{q}{\beta(2\pi)^3}\int_{\mathbb{R}^3} \log\left(1+e^{\beta\mu-\beta|k|^2}\right)\,\mathrm{d}k, \qquad q = \begin{cases} 2 & \text{spin-1/2,} \\ 1 & \text{spin-polarized.} \end{cases}$$

Theorem (Spin-1/2 (Seiringer 2006))

Let $v \geq 0$ be radial and of compact support. Then for small $a_s^3\rho$ uniformly in $z = e^{\beta\mu} \gtrsim 1$

$$\psi(\beta,\mu) = \psi_0(\beta,\mu) - 2\pi a_s \rho_0^2[1 + o(1)].$$

Theorem (Spin-polarized (L.–Seiringer 2023))

Let $v \geq 0$ be radial and of compact support. Then for small $a_p^3\rho$ uniformly in $z = e^{\beta\mu} \gtrsim 1$

$$\psi(\beta,\mu) \geq \psi_0(\beta,\mu) - 24\pi \frac{-\operatorname{Li}_{5/2}(-z)}{(-\operatorname{Li}_{3/2}(-z))^{5/2}} a_p^3 \rho_0^{8/3}[1 + o(1)].$$

$\rho = \partial_\mu \psi(\beta,\mu)$ is the density and $\rho_0 = \partial_\mu \psi_0(\beta,\mu)$ is the density of a free gas.

$$-\operatorname{Li}_s(-e^x) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1}}{e^{t-x}+1}\,\mathrm{d}t$$

(Analogous results in dimensions $d = 1, 2$.)

Cluster expansion (spin-polarized case)

Way of calculating $\rho^{(n)}_{\text{Jas}}$. (Formal calculations of Gaudin–Gillespie–Ripka 1971.)

Recall

$$C_N = \int \cdots \int \prod_{1 \le i < j \le N} f(x_i - x_j)^2 |D_N(x_1, \ldots, x_N)|^2 \, \mathrm{d}x_1 \ldots \mathrm{d}x_N$$

Introduce for $e = (i, j)$

$$g(x) = f(x)^2 - 1, \qquad g_e = g_{ij} = g(x_i - x_j) = f(x_i - x_j)^2 - 1,$$

$$D_N(x_1, \ldots, x_N) = \det[u_k(x_i)]_{\substack{1 \le i \le N \\ k \in M_F}}, \qquad |D_N|^2 = \det[\gamma^{(1)}(x_i; x_j)]_{1 \le i, j \le N},$$

with M_F some set of momenta (think $M_F \simeq B_F$ the Fermi ball), and

$$\gamma_{ij} = \gamma^{(1)}(x_i; x_j) = \frac{1}{L^3} \sum_{k \in M_F} e^{ik(x_i - x_j)}$$

the one-particle density matrix of ψ_N.

Define $\mathcal{G}_p$: Set of all graphs on p vertices with all vertices degree ≥ 1. Then

$$C_N = \int \cdots \int \prod_{e \in \text{Complete graph}} (1 + g_e) \det[\gamma_{ij}]_{i,j \leq N} \, \mathrm{d}x_1 \ldots \mathrm{d}x_N$$

$$= \int \cdots \int \left[1 + \sum_{p=2}^{N} \binom{N}{p} \sum_{G \in \mathcal{G}_p} \prod_{e \in G} g_e \right] \det[\gamma_{ij}]_{i,j \leq N} \, \mathrm{d}x_1 \ldots \mathrm{d}x_N$$

$$= N! \left[1 + \sum_{p=2}^{N} \frac{1}{p!} \int \cdots \int \sum_{G \in \mathcal{G}_p} \prod_{e \in G} g_e \det[\gamma_{ij}]_{i,j \leq p} \, \mathrm{d}x_1 \ldots \mathrm{d}x_p \right]$$

$$= N! \left[1 + \sum_{p=2}^{\infty} \frac{1}{p!} \int \cdots \int \sum_{G \in \mathcal{G}_p} \prod_{e \in G} g_e \det[\gamma_{ij}]_{i,j \leq p} \, \mathrm{d}x_1 \ldots \mathrm{d}x_p \right].$$

Use Leibniz formula: $\det[\gamma_{ij}]_{1 \leq i,j \leq p} = \sum_{\pi \in \mathcal{S}_p} (-1)^{\pi} \gamma_{1\pi(1)} \cdots \gamma_{p\pi(p)}$:

$$= N! \left[1 + \sum_{p=2}^{\infty} \frac{1}{p!} \sum_{G \in \mathcal{G}_p} \sum_{\pi \in \mathcal{S}_p} (-1)^{\pi} \int \cdots \int \prod_{e \in G} g_e \prod_{j=1}^{p} \gamma_{j\pi(j)} \, \mathrm{d}x_1 \ldots \mathrm{d}x_p \right].$$

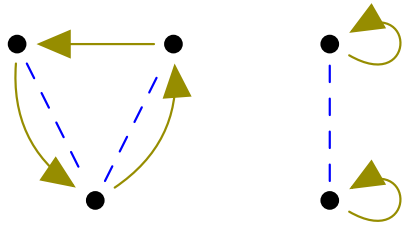
Definition (Diagrams)

- $\mathcal{G}_p$: Set of all graphs on p vertices with all vertices degree ≥ 1.
- A **diagram** (π, G) (on p vertices) is a pair of a permutation $\pi \in \mathcal{S}_p$ (viewed as a directed graph) and a graph $G \in \mathcal{G}_p$.
- $\mathcal{D}_p$: Set of all diagrams (on p vertices).
- Value of $(\pi, G) \in \mathcal{D}_p$ is

$$\Gamma_{\pi,G} := (-1)^{\pi} \int \cdots \int \prod_{e \in G} g_e \prod_{j=1}^{p} \gamma_{j\pi(j)} \, \mathrm{d}x_1 \ldots \mathrm{d}x_p.$$

- A diagram (π, G) is **linked** if $\pi \cup G$ is connected.
- $\mathcal{L}_p$: Set of all linked diagrams (on p vertices).

$$\frac{C_N}{N!} = 1 + \sum_{p=2}^{\infty} \frac{1}{p!} \sum_{(\pi,G) \in \mathcal{D}_p} \Gamma_{\pi,G}.$$



$$= \int \cdots \int g_{12} g_{23} g_{45} \gamma_{12} \gamma_{23} \gamma_{31} \gamma_{44} \gamma_{55}$$

Decomposing into linked components ($\sim$ "standard cluster expansion")

$$\frac{C_N}{N!} = 1 + \sum_{p=2}^{\infty} \frac{1}{p!} \sum_{(\pi,G)\in\mathcal{D}_p} \Gamma_{\pi,G} = \exp\left[\sum_{p=2}^{\infty} \frac{1}{p!} \sum_{(\pi,G)\in\mathcal{L}_p} \Gamma_{\pi,G}\right].$$

Absolute convergence:

$$\sum_{p=2}^{\infty} \frac{1}{p!} \left| \sum_{(\pi,G)\in\mathcal{L}_p} \Gamma_{\pi,G} \right| < \infty.$$

Combinatorics: reduces to

$$\rho \int |g(x)|\,\mathrm{d}x \int |\gamma^{(1)}(x;0)|\,\mathrm{d}x$$
$$= \rho \times a_p^3 \left|\log a_p^3 \rho\right| \times \frac{1}{L^3} \int_{[0,L]^3} \left| \sum_{k\in M_F} e^{ikx} \right| \mathrm{d}x < c.$$

(Same condition for convergence of formula for $\rho^{(2)}_{\mathrm{Jas}}$.)

Absolute convergence

Holds if $$a_p^3\rho\left|\log a_p^3\rho\right| \times \frac{1}{L^3}\int_{[0,L]^3}\left|\sum_{k\in M_F} e^{ikx}\right| \mathrm{d}x < c.$$

Introduce Fermi polyhedron:

For Fermi ball B_F and Fermi polyhedron P_F (Kolomoitsev–Lomako 2018):

$$\frac{1}{L^3}\int_{[0,L]^3}\left|\sum_{k\in B_F} e^{ikx}\right| \mathrm{d}x \sim N^{1/3}, \qquad \frac{1}{L^3}\int_{[0,L]^3}\left|\sum_{k\in P_F} e^{ikx}\right| \mathrm{d}x \lesssim s(\log N)^3.$$

s is number of corners of P_F – some parameter to choose.

- Finite size error from kinetic energy $\sim N^{-1/3}\rho^{5/3}$.
- Leading energy density from interaction $\sim a_p^3\rho^{8/3}$.

$\rightsquigarrow$ Need $N^{-1/3}\rho^{5/3} \ll a_p^3\rho^{8/3}$, i.e. $N^{1/3}a_p^3\rho \gg 1$.

$\rightsquigarrow$ **Can't use Fermi ball.**

Calculation of energy

One concludes

$$\rho^{(2)}_{\mathrm{Jas}}(x_1, x_2) = f(x_1 - x_2)^2 \left[\rho^{(2)}(x_1, x_2) + \varepsilon^{(2)}_{\mathrm{Jas}}(x_1, x_2)\right]$$

with explicit error $\varepsilon^{(2)}_{\mathrm{Jas}}$.

$\rightsquigarrow$ Energy is

$$\begin{aligned}&\langle \psi_N | H_N | \psi_N \rangle \\ &\simeq \sum_{k \in P_F} |k|^2 + L^3 \int \left[\rho^{(2)}(x, 0) + \varepsilon^{(2)}_{\mathrm{Jas}}(x, 0)\right] \left[|\nabla f(x)|^2 + \frac{1}{2} v(x) f(x)\right] \mathrm{d}x.\end{aligned}$$

Calculation of energy

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$\rightsquigarrow$ Energy is

$$\begin{aligned}&\langle \psi_N | H_N | \psi_N \rangle \\ &\simeq \sum_{k \in P_F} |k|^2 + L^3 \int \left[\rho^{(2)}(x, 0) + \varepsilon^{(2)}_{\mathrm{Jas}}(x, 0)\right] \left[|\nabla f(x)|^2 + \frac{1}{2} v(x) f(x)\right] \mathrm{d}x.\end{aligned}$$

Taylor expand $\varepsilon^{(2)}_{\mathrm{Jas}} \rightsquigarrow$ need bounds on

$$\frac{1}{L^3} \int_{[0,L]^3} \left| \sum_{k \in P_F} k^\mu e^{ikx} \right| \mathrm{d}x, \qquad \frac{1}{L^3} \int_{[0,L]^3} \left| \sum_{k \in P_F} k^\mu k^\nu e^{ikx} \right| \mathrm{d}x$$

$\rightsquigarrow$ Choose P_F in technical way. Long/Technical computation based on (Kolomoitsev–Lomako 2018) involving (in particular) some elementary number theory.

Calculating Taylor expansion more precisely

$$\rho^{(2)}_{\mathrm{Jas}}(x_1, x_2) \simeq \frac{(6\pi^2)^{2/3}}{5} f(x_1 - x_2)^2 \rho^{8/3} |x_1 - x_2|^2 \left[1 - \frac{3(6\pi^2)^{2/3}}{35} \rho^{2/3} |x_1 - x_2|^2\right].$$

Thus

$$\begin{aligned}
&\langle \psi_N | H_N | \psi_N \rangle \\
&\quad \simeq \sum_{k \in P_F} |k|^2 + \frac{(6\pi^2)^{2/3}}{10} L^3 \rho^{8/3} \int \mathrm{d}x \left[|x|^2 - \frac{3(6\pi^2)^{2/3}}{35} \rho^{2/3} |x|^4\right] \\
&\qquad \times \left[|\nabla f(x)|^2 + \frac{1}{2} v(x) f(x)\right] \\
&\quad \simeq \frac{3}{5}(6\pi^2)^{2/3} L^3 \rho^{5/3} + \frac{12\pi}{5}(6\pi^2)^{2/3} L^3 a_p^3 \rho^{8/3} \left[1 - \frac{9}{35}(6\pi^2)^{2/3} a_0^2 \rho^{2/3}\right]
\end{aligned}$$

Thanks for your attention!

- A. B. Lauritsen. "Almost optimal upper bound for the ground state energy of a dilute Fermi gas via cluster expansion". arXiv:2301.08005 [math-ph]. 2023.
- A. B. Lauritsen and R. Seiringer. "Ground state energy of the dilute spin-polarized Fermi gas: Upper bound via cluster expansion". arXiv:2301.04894 [math-ph]. 2023.
- A. B. Lauritsen and R. Seiringer. "Pressure of a dilute spin-polarized Fermi gas: Lower bound". arXiv:2307.01113 [math-ph]. 2023.